SOFTWARE ENGINEERING LAB 5

Static Code Analysis

Flight Reservation System

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1. Objective and Deliverables

The objective of this lab is to use static-analysis tools to identify and correct code-quality, security, reliability, and style issues in **flight_reservation_system.py**.

2. Before Corrections: Static Analysis

The original Flight Reservation System source was scanned before the cleanup. The baseline results recorded during the lab showed a Pylint score of 4.00/10. Bandit reported security findings, including high-severity issues. Flake8 identified style and formatting problems.

Tool	Before result	Interpretation
Pylint	4.00/10	Code-quality and maintainability issues were present.
Bandit	2 high, 1 medium, 3 low findings (recorded baseline)	Potentially unsafe patterns needed review and remediation.
Flake8	Multiple findings	Style, naming, whitespace, and other lint issues needed correction.

3. Issue Table

The following table summarizes the main issues addressed in the corrected source. Line numbers can change after edits, so verify exact line numbers against the original file if required by the instructor.

Issue	Category	Fix applied
Mutable default argument	Reliability	Changed the default to None and initialized a new list inside the function.
Unsafe eval() use	Security	Replaced expression evaluation with explicit parameters/safer handling; removed execution of arbitrary expression strings.
Shell command execution	Security	Removed os.system-based output and used ordinary Python file/output operations.
Hardcoded secret	Security	Removed the hardcoded secret and stopped printing it.
Weak MD5 hash	Security	Replaced MD5 with SHA-256 for the passenger fingerprint.
Predictable random PIN	Security	Used secrets.randbelow() to generate the boarding PIN.
Bare exception handling	Reliability	Replaced broad exception handling with an explicit booking-membership check.
Naming, documentation, and formatting	Maintainability/style	Applied snake_case naming, docstrings, and autopep8 formatting.
Unused imports / variables	Maintainability	Removed unused items identified during cleanup.

4. After Corrections: Changes Made

The corrected version focuses on removing unsafe patterns, improving reliability, and making the source easier to read and maintain.

Before	After	Reason
Mutable default list	None default; create list inside function	Prevents state being shared across calls.
eval() for surcharge/formula	Explicit surcharge-rate parameter / safer calculation	Avoids arbitrary expression execution.
os.system() to display/save information	Python file handling and print()	Avoids shell-command injection risks.
Hardcoded secret printed to terminal	Secret removed from source/output	Prevents accidental secret exposure.
hashlib.md5()	hashlib.sha256()	Uses a stronger hash function.
random-based PIN	secrets.randbelow()	Uses a security-oriented random source.
Bare except	Explicit condition checking	Makes failure behavior clearer and avoids swallowing errors.
Inconsistent naming / formatting	snake_case, docstrings, autopep8	Improves readability and tool compliance.

A formatting pass was applied with autopep8. The final source was run to check that the demonstration still executes and produces a boarding PIN, passenger hash, baggage fee, flight availability output, and manifest confirmation.

5. After Corrections: Verification Results

The final reports and program run recorded during the lab show the following results.

Tool / check	Before	After
Pylint	4.00/10	10.00/10
Bandit	2 high, 1 medium, 3 low findings	0 reported issues
Flake8	Multiple style findings	No output (clean run)
Python execution	Ran with problematic patterns	Program ran and printed expected demo outputs

6.Before — Pylint

```
PS C:\Users\SHREYAS\Documents\SE_LAB_PES2UG24CS490\lab5\before> & "$HOME\Documents\L5_50_flight_reservation_system\venv\Scripts\python.exe" -m pylint .\flig
ht_reservation_system.py
************* Module flight_reservation_system
flight_reservation_system.py:23:0: C0301: Line too long (132/100) (line-too-long)
flight_reservation_system.py:87:0: C0304: Final newline missing (missing-final-newline)
flight_reservation_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
flight_reservation_system.py:12:0: C0103: Constant name "total_revenue" doesn't conform to UPPER_CASE naming style (invalid-name)
flight_reservation_system.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:15:0: C0103: Function name "bookFlightSeat" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:15:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
flight_reservation_system.py:16:4: W0603: Using the global statement (global-statement)
flight_reservation_system.py:23:24: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
flight_reservation_system.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:27:0: C0103: Function name "cancelBooking" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:31:4: W0702: No exception type(s) specified (bare-except)
flight_reservation_system.py:35:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:35:0: C0103: Function name "checkSeatAvailability" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:37:7: C0121: Comparison 'flights.get(flight_no) == None' should be 'flights.get(flight_no) is None' (singleton-comparison)
flight_reservation_system.py:39:4: R1703: The if statement can be replaced with 'return bool(test)' (simplifiable-if-statement)
flight_reservation_system.py:39:4: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
flight_reservation_system.py:36:4: W0612: Unused variable 'airport_fee' (unused-variable)
flight_reservation_system.py:45:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:45:0: C0103: Function name "calculateBaggageFee" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:47:16: W0123: Use of eval (eval-used)
flight_reservation_system.py:46:4: W0612: Unused variable 'base_fare' (unused-variable)
flight_reservation_system.py:51:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:51:0: C0103: Function name "generatePassengerHash" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:55:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:55:0: C0103: Function name "generateBoardingPin" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:59:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:59:0: C0103: Function name "exportFlightManifest" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:60:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
flight_reservation_system.py:60:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
flight_reservation_system.py:66:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:66:0: C0103: Function name "getFlightSummary" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:66:21: W0621: Redefining name 'flights' from outer scope (line 6) (redefined-outer-name)
flight_reservation_system.py:68:8: W0612: Unused variable 'f' (unused-variable)
flight_reservation_system.py:74:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:4:0: W0611: Unused import math (unused-import)
```

7.Before — Flake8 and Bandit summary

```
--------------------------------------------------

Code scanned:
        Total lines of code: 65
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 3
                Medium: 1
                High: 2
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 5
Files skipped (0):
```

```
PS C:\Users\SHREYAS\Documents\SE_LAB_PES2UG24CS490\lab5\before> & "$HOME\Documents\L5_50_flight_reservation_system\venv\Scripts\python.exe" -m flake8 .\flig
ht_reservation_system.py
.\flight_reservation_system.py:4:1: F401 'math' imported but unused
.\flight_reservation_system.py:7:80: E501 line too long (84 > 79 characters)
.\flight_reservation_system.py:8:80: E501 line too long (85 > 79 characters)
.\flight_reservation_system.py:9:80: E501 line too long (83 > 79 characters)
.\flight_reservation_system.py:23:80: E501 line too long (132 > 79 characters)
.\flight_reservation_system.py:31:5: E722 do not use bare 'except'
.\flight_reservation_system.py:36:5: F841 local variable 'airport_fee' is assigned to but never used
.\flight_reservation_system.py:37:31: E711 comparison to None should be 'if cond is None:'
.\flight_reservation_system.py:46:5: F841 local variable 'base_fare' is assigned to but never used
.\flight_reservation_system.py:87:7: W292 no newline at end of file
```

After — Program execution and Pylint

```
PS C:\Users\SHREYAS\Documents\SE_LAB_PES2UG24CS490\lab5\after> & "$HOME\Documents\L5_50_flight_reservation_system\venv\Scripts\python.exe" .\flight_reservat
ion_system.py
Is AI101 open: True
Baggage Fee: 67.5
Passenger Hash: 76161a689c7529645b02678017fed8b94a047e8aa458a82c50a23924f445ee07
Boarding PIN: 8540
Boarding manifest saved to flight_manifest.txt
Available flights: 3
```

After — Bandit

```
PS C:\Users\SHREYAS\Documents\SE_LAB_PES2UG24CS490\lab5\after> & "$HOME\Documents\L5_50_flight_reservation_system\venv\Scripts\python.exe" -m bandit .\fligh
t_reservation_system.py
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.12.10
Run started:2026-10-14 15:50:27.316258+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 88
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

After — Flake8

```
Files skipped (0):
PS C:\Users\SHREYAS\Documents\SE_LAB_PES2UG24CS490\lab5\after> & "$HOME\Documents\L5_50_flight_reservation_system\venv\Scripts\python.exe" -m flake8 .\fligh
t_reservation_system.py
PS C:\Users\SHREYAS\Documents\SE_LAB_PES2UG24CS490\lab5\after>
```